Dario Napolitano

Computer Science - Software Engineer
LinkedIn, GitHub, napolitanodar.io, darionapo2.0@gmail.com, 20124 Milan Italy, +39 388 259 1181

Professional Experience

Moviri Consulting, Performance Engineering, Milan Italy, 12/2024 - current
- Designed a scalable ETL pipeline, streaming Dynatrace RUM data into Splunk, effectively automating data analysis and reporting.
- Designed custom load testing scripts to benchmark the performance impact of a corporate antivirus solution in the banking industry.
- Instrumented and configured full infrastructure observability with Dynatrace.
- Developed Selenium endurance load tests to benchmark and collect user experience data from complex applications.

Weflex Italia, Technical Office, Modena Italy, 08/2019 - 12/2024
- Developed warehouse logistics layouts for more than 20 customers in the ceramic tile industry.
- Used real-time tracking systems and ERP software to improve merchandise management and movement.
- Successfully took part in international projects in the US.

searchingpatents.com, Full Stack Developer, Bologna Italy, 10/2022 - 09/2025
- Built a custom Flask app for retrieval of patents data from European Patent Office.
- Implemented automated report generation according to selectable preferences, decreasing report time by 90%.
- Released multiple application versions, continuously improving workflow efficiency.

Education

MSE, Politecnico of Milan, Computer Science and Engineering, 09/2024 - current
- Coursework: Deep Learning, SWE, Compilers, NLP, Recommender Systems, Dynamic Programming, Control Systems

BSE, University of Modena and Reggio Emilia, Computer Engineering, 09/2021 - 10/2024
- Coursework: Linear Algebra, Physics, Computer Architecture, OOP, Telecommunications, IoT, Computer Vision
- Erasmus+ mobility at Loyola University, Seville Spain

Projects

Arduino Hackathon, @POLIMI NECTSLab, repo
Built an Arduino-based home safety system, combining voice keyword spotting with a local real-time anomaly detection model (Isolation Forest) to detect habit disruptions. Integrated remote-control capabilities and automated alert delivery via web dashboards.
- Tools used: Arduino UNO Q, scikit-learn, Firebase

Deep Learning Competitions, @POLIMI, repo
Tackled two vision challenges: blood cell image classification and Martian terrain semantic segmentation. Applied transfer learning and fine tuning to CNNs and U-Nets achieving up to +20% accuracy boost and securing a top 10 leaderboard placement. Designed and optimized ensemble strategies that successfully improved overall accuracy and mIoU. Implemented data augmentation, class rebalancing, and loss function tuning to mitigate overfitting and enhance rare-class performance.
- Tools used: TensorFlow, PyTorch, Keras, OpenCV, scikit-learn

Visual Impairment Assistant, (Thesis project) @UNIMORE ARSControl, repo
Developed a computer vision prototype to assist visually impaired individuals in navigating urban and domestic environments. Integrated real-time stereo vision depth estimation and object detection on Raspberry Pi 4.
- Tools used: OpenCV, TensorFlow, INTEL RealSense, multiprocessing

Natural Language Processing competition, @POLIMI, repo
Built an end-to-end multimodal AI pipeline on a large medical imaging dataset, covering data analysis, text-based image retrieval, and classification. Fine-tuned CNNs and multimodal language models (MedGemma, Qwen 2.5-VL) for image captioning and analysis. Built a semantic search engine using image embeddings.
- Tools used: PyTorch, TensorFlow, Keras, Hugging Face Transformers, CLIP, scikit-learn

Flutter application, @POLIMI, repo
Designed and developed a Flutter mobile application for pantry tracking and smart recipe discovery, integrating a Firebase backend with Gemini AI and Spoonacular APIs for retrieval of recipes suggestions. Conducted extensive unit and integration testing.
- Tools used: Flutter, Gemini API, Firebase

Particulate Matter Detection, (school project) @ISAC/CNR
Programmed an ESP32 microcontroller for environmental data detection, monitoring air quality parameters in real time. Integrated real time data transmission protocols. Designed a relational database architecture. Developed and deployed a web platform for visualizing telemetry data, fully meeting project requirements.
- Tools used: C++, LoRa, MQTT, PostgreSQL, Flask, Angular

Skills

- Programming: Python, JavaScript, C/C++, Java, Bash, SQL, MATLAB.
- AI and Data: NumPy, pandas, scikit-learn, TensorFlow, Keras, OpenCV.
- DevOps: Kubernetes, Docker, AWS, Terraform.
- Observability and Performance testing: OpenTelemetry, Splunk, Dynatrace, Grafana, K6, NeoLoad.
- Frameworks: Angular, Flask, Flutter, React, Django.
- Certifications: Dynatrace Associate
- Spoken languages: **English**: **C1** IELTS overall 8/9, **Spanish**: **B2**, **Italian**: Native Speaker.